ENVS 193DS Infographic

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Set up

```r
# loaded in packages
library(tidyverse)
library(janitor)
library(here)
library(webr)
library(ggimage)
library(showtext)

library(dplyr)
library(ggplot2)

# read in data and saved as object called `rxntime`
rxntime   <- readr::read_csv(here::here("data","new-data.csv"))

# read in my data and saved as object called 'pie'
pie <- readr::read_csv(here::here("data","pie-data.csv"))

# imported fonts
# 1st argument: google name,
# 2nd name: font name in R
font_add_google('Exo 2', 'exo2', 700)
font_add_google('Exo 2', 'exo')

# enabled showtext font rendering
showtext_auto()
```

Part 3. Visualizing data

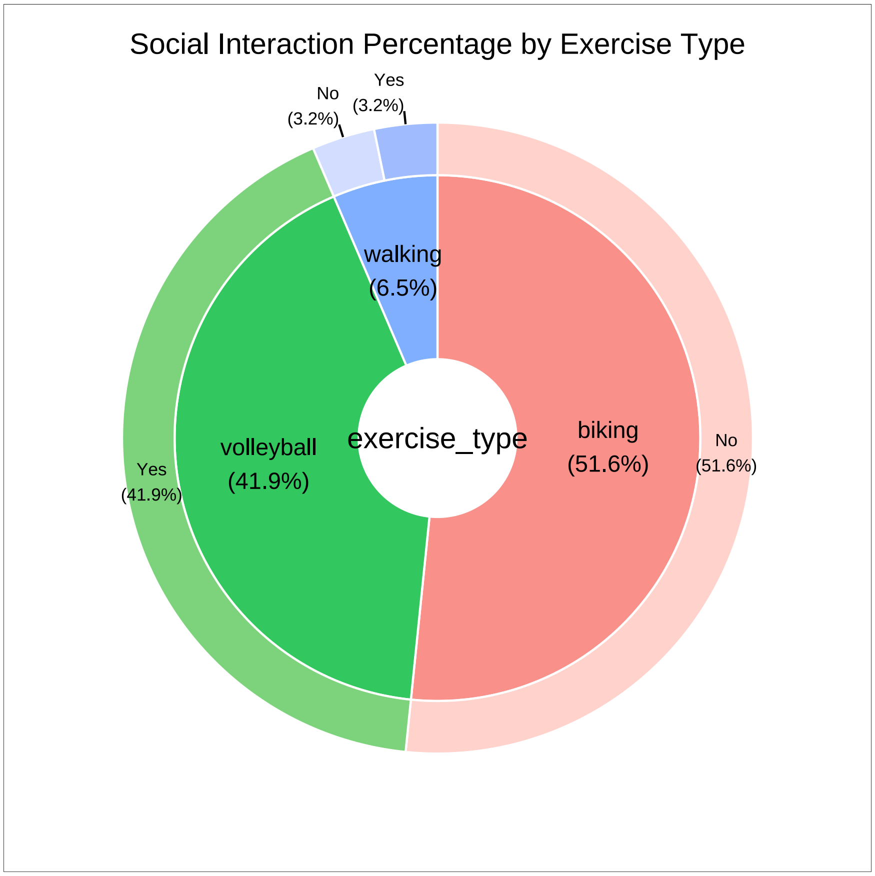
Pie chart

Cleaning data

```r
# created a new object from pie
pie_clean <- pie |>
  # cleaned column names
  clean_names() |>
  # selected columns of interest
  select(exercise_type, social_interaction, count) |>
  # changed existing abbreviations
  mutate(social_interaction = case_when(
    social_interaction == "y" ~ "Yes",
    social_interaction == "n" ~ "No"
  )) |>
  # removed missing values
  drop_na()
```

Visualization

```r
# save plot as an object <- created a pie donut chart with pie_clean df
piechart <- PieDonut(data = pie_clean,
         # pie slices by exercise type, outlined by social interaction
         aes(exercise_type, social_interaction, count = count),
         r0 = 0.3, # inner ring thickness
         r1 = 1, # outer ring thickness
         ratioByGroup = FALSE, # outer circle of the pie adds up to 100%
         title = "Social Interaction Percentage by Exercise Type") # title
```




```
# PNG export
ggsave("C:/github/ENVS-193DS_spring-2026_infographic/exports/pie.png",
       plot = get_last_plot(), # selected plot to save
       device = ragg::agg_png, # supports alpha
       bg = "transparent", # transparent background
       width = 4, # adjusted width
       height = 3, # adjusted height
       units = "in", # units of plot measurements
       dpi = 300) # resolution
```

Scatter plot

Cleaning data

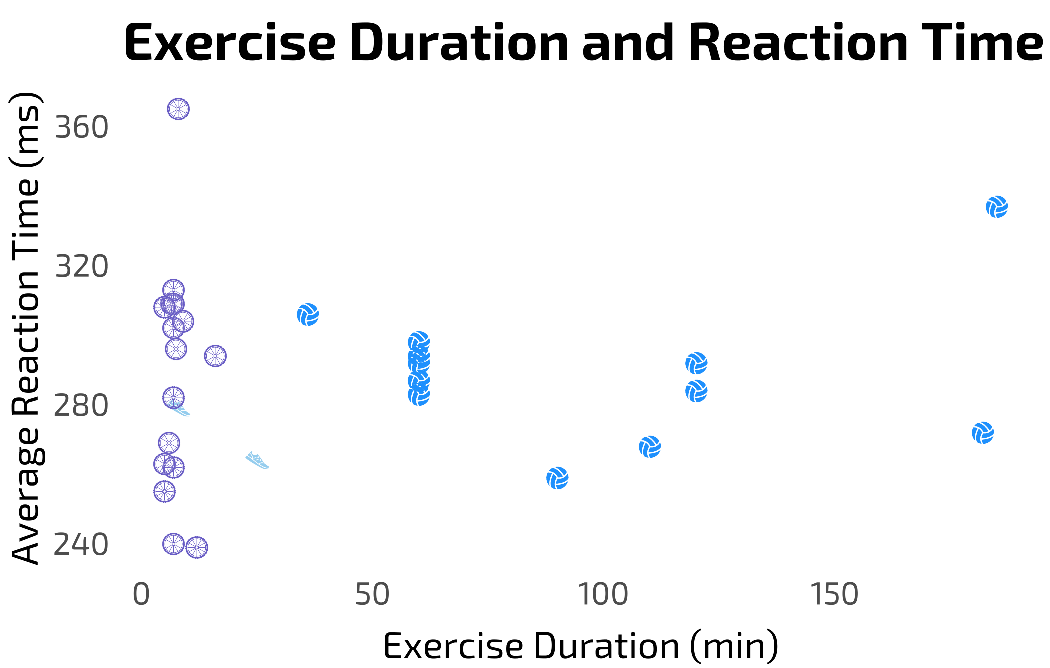
```r
# created a new object from rxntime
rxntime_clean <- rxntime |>
  # cleaned column names
  clean_names() |>
  # selected columns of interest
  select(exercise_type, exercise_duration_min, reaction_time_avg_ms) |>
  # removed missing values
  drop_na()
```

Visualization

```r
# created image map by exercise type
img_map <- c(
  biking = "C:/github/ENVS-193DS_spring-2026_infographic/images/biking.png",
  volleyball = "C:/github/ENVS-193DS_spring-2026_infographic/images/vb.png",
  walking = "C:/github/ENVS-193DS_spring-2026_infographic/images/walking.png"
)

# new object from clean data frame
rxntime_scatter <- rxntime_clean |>
  # mutated data frame for the addition of icons
  mutate(exercise_type = as.character(exercise_type),
         image = img_map[exercise_type])
```

```r
# called showtext
showtext_auto()

# saved plot as an object <- base layer: ggplot
scatterplot <- ggplot(rxntime_scatter,
            # x-axis: exercise duration in minutes
            # y-axis: avg reaction time in milliseconds
            aes(x = exercise_duration_min,
                y = reaction_time_avg_ms)) +
  # first layer: data points represented by custom icons
  geom_image(aes(
    image = image), # icon placement
    size = 0.05, # icon size
    asp = 1) + # square aspect ratio
  # custom theme
  theme_minimal() +
```

```
  theme(
    legend.position = "none", # removed legend
    panel.background = element_rect(fill = NA, # removed fill and color
                                    color = NA),
    panel.grid = element_blank(), # removed panel grid
    axis.ticks = element_blank(), # removed axis ticks
    plot.background = element_rect(fill = NA, # removed fill and color
                                  color = NA),
    plot.title = element_text(family = "exo2", # title custom font
                             size = 20),
    axis.title = element_text(family = "exo", # axis title font
                             size = 14),
    axis.title.x = element_text(margin = margin(t = 8)), # text spacing
    axis.text = element_text(family = "exo", size = 12)) + # tick label
  # relabeled x- and y- axes, adding title
  labs(title = "Exercise Duration and Reaction Time",
       x = "Exercise Duration (min)",
       y = "Average Reaction Time (ms)")

# view figure in plots
print(scatterplot)
```



```r
# PNG export
ggsave("C:/github/ENVS-193DS_spring-2026_infographic/exports/scatterplot.png",
       plot = get_last_plot(), # selected plot to save
       device = ragg::agg_png, # supports alpha
       bg = "transparent", # transparent background
       width = 4, # adjusted width
       height = 3, # adjusted height
       units = "in", # units of plot measurements
       dpi = 300) # resolution
```

Box / jitter plot

Cleaning data

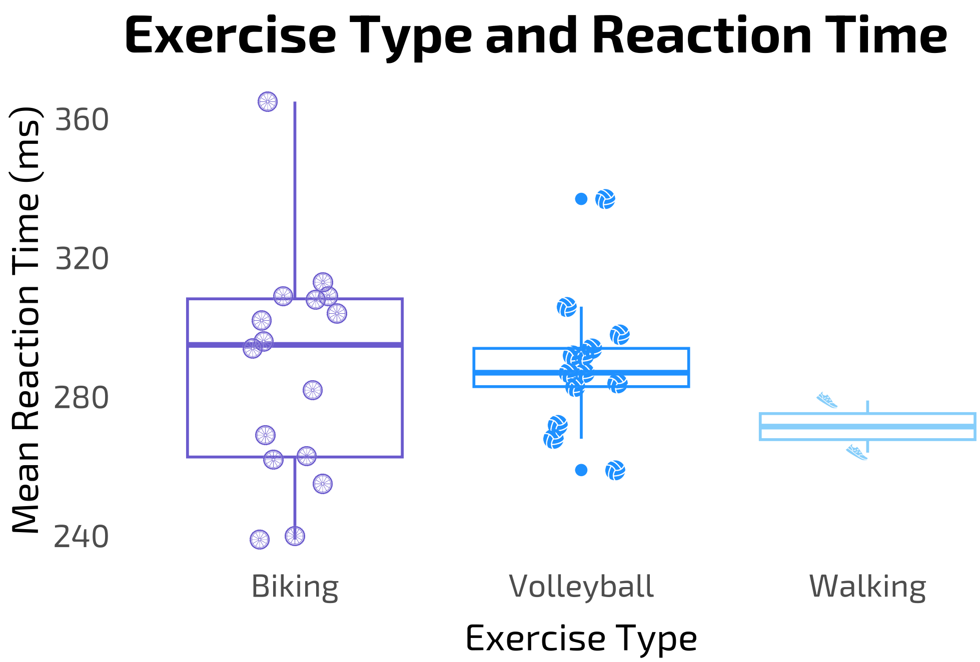
```r
# created a new object from rxntime
rxntime_box <- rxntime |>
  # cleaned column names
  clean_names() |>
  # selected columns of interest
  select(exercise_type, reaction_time_avg_ms) |>
  # created a new column for images in df
  mutate(image = img_map[exercise_type]) |>
  # capitalized existing abbreviations
  mutate(exercise_type = case_when(
    exercise_type == "volleyball" ~ "Volleyball",
    exercise_type == "biking"    ~ "Biking",
    exercise_type == "walking"   ~ "Walking",
    # assigned exercise types to images
    TRUE ~ as.character(exercise_type)
  )) |>
  # removed missing values
  drop_na()
```

Visualization

```r
# called showtext
showtext_auto()
```

```r
# save plot as an object <- base layer: ggplot
boxplot <- ggplot(data = rxntime_box,
       # x-axis: exercise type
       # y-axis: mean reaction time
       # color by exercise type
       mapping = aes(x = exercise_type,
                     y = reaction_time_avg_ms,
                     color = exercise_type)) +
  # first layer: boxplot
  geom_boxplot() +
  # second layer: icons and jitter
  geom_image(
    aes(image = image),
    position = position_jitter(width = 0.15,
                               height = 0),
    size = 0.045, # icon size
    asp = 1) + # icon aspect ratio
  # added title and relabeled x- and y-axes
  labs(title = "Exercise Type and Reaction Time",
       x = "Exercise Type",
       y = "Mean Reaction Time (ms)") +
  # custom theme
  theme_minimal() +
  # removed legend
  theme(legend.position = "none", # removed legend
        panel.background = element_rect(fill = NA, # removed fill and color
                                        color = NA),
        panel.grid = element_blank(), # removed panel grid
        axis.ticks = element_blank(), # removed axis ticks
        plot.background = element_rect(fill = NA, # removed fill and color
                                      color = NA),
        plot.title = element_text(family = "exo2", # title custom font
                                  size = 20),
        axis.title = element_text(family = "exo", # axis title font
                                  size = 14),
        axis.title.x = element_text(margin = margin(t = 8)), # text spacing
        axis.text = element_text(family = "exo", size = 12)) + # tick label font
  # custom colors
  scale_color_manual(values = c("Biking" = "slateblue",
                                "Volleyball" = "dodgerblue",
                                "Walking" = "lightskyblue")) +
  # removed grid and axis lines
```

```r
  theme(panel.grid = element_blank(),
        axis.line = element_blank())

# view figure in plots
print(boxplot)
```




```r
# PNG export
ggsave("C:/github/ENVS-193DS_spring-2026_infographic/exports/box.png",
       plot = get_last_plot(), # selected plot to save
       device = ragg::agg_png, # supports alpha
       bg = "transparent", # transparent background
       width = 4, # adjusted width
       height = 3, # adjusted height
       units = "in", # units of plot measurements
       dpi = 300) # resolution
```

Part 4. Write about your process

General design and visualization information

The donut pie chart was constructed around the frequency of each exercise type, but I also attempted to include information about socialization in the outer ring of the chart. I positioned it towards the top because it seemed like the easiest to understand visually and would make a good introduction to the exercise types referenced in the other figures. The scatter plot highlights the relationship between exercise duration and reaction time, and the box plot indicates the relationship between exercise type and reaction time. I wanted to show that there was a great variety in reaction times across exercise sessions of different time lengths for the former, and that mean reaction times ended up being different across all the exercise types for the latter.

At first, I thought of leaning into a "digital" theme because data input happened via my phone, the Google Sheets app, and an online reaction time test. But my initial choices in fonts were rigid and blocky. Paired with darker color palettes, they looked uninteresting and gloomy.

I went with blue because I associate it with vitality. Energy is needed for exercise and exercising revitalizes. I also associate the color with the sky here on nice days. Those are the best days to bike, walk, and play volleyball. A more relaxed font was used in conjunction with Exo and Exo 2 for a mix of words that felt balanced and easy on the eyes.

Sources and process

The donut pie chart was inspired by The Data Digest, a YouTuber who uses statistical analysis to discuss Magnus Carlsen's chess matches, TV show endings, and other topics. He created a tutorial that I referenced to create my own pie donut.

The box and scatter plots were visualizations we've become familiar with in class, but the icons were inspired by an affective visualization I saw in our last section on Friday. A classmate had used birds to represent raw data points, and the buildings on which the birds sat also represented their findings. I wanted to try something similar, at the very least by adding icons to my figures. For both plots, I cleaned the data and used geom_image to place imported icons. The box plot required geom_boxplot.

I troubleshooted errors in my code through forums on StackOverflow (others had run into the same issues I did and sought help from experts), watched tutorials on YouTube, and looked up guides on Google (mostly for showtext and the webr pie donut) in attempts to make the code work.